Mahriar Gharaghani
AI / ML Engineer | Python Developer | Decision Intelligence (AI & OR)

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Profile

AI/ML engineer with hands-on experience across full-stack applications, workflow automation, decision-support tools, and applied AI. Research focus includes sequential decision-making under uncertainty and optimization-informed AI systems, motivated by real project portfolio budgeting constraints.

Research Interests

Reinforcement learning for operations; sequential decision-making under uncertainty; optimization and simulation for project portfolio management; cash-flow constrained budgeting; decision intelligence; applied RAG/LLM systems for domain workflows.

Education

M.Sc. in Engineering Management 2023–2026 (in progress)
Iran University of Science and Technology (IUST), Tehran, Iran
Thesis (working title): *Autonomous Budgeting AI Agent / Developing an AI Agent for Project Portfolio Budget Allocation under Uncertainty and Dynamic Cash-Flow Constraints*

- Built a Python-based deep reinforcement learning system for project portfolio budget allocation under uncertainty and dynamic financial constraints.
- Evaluation compares learned policies to a perfect-foresight optimization benchmark (upper bound).
- Thesis context includes external support under a contract with **Tehran Industrial Estates Company**.

B.Sc. in Industrial Engineering 2017–2023
Isfahan University of Technology (IUT), Isfahan, Iran

Ph.D. Entrance Exam (Computer Engineering — AI & Robotics) 2026 (results pending)
National Universities of Iran

- Built **RAG4Konkur**: a local AI study system for flashcard generation, concept review, and AnkiDroid-ready exports.
- Documented preparation topics and artifacts on GitHub (data structures & algorithms, machine learning, pattern recognition).

Research Experience

M.Sc. Thesis Researcher, IUST 2023–2026

- Designed an AI-driven sequential decision-making framework for multi-period portfolio budget allocation under uncertain and dynamic cash-flow constraints.
- Integrated optimization, simulation, and policy learning to produce feasible allocation decisions under uncertainty.

- Implemented end-to-end experimentation in Python (environment, training loop, evaluation, and reporting).
- Benchmarking: compared learned policies against a perfect-foresight optimization solution as an upper bound reference.

Industry-Supported Thesis Context (Tehran Industrial Estates Company) 2023–2026

- Aligned the problem formulation with real institutional needs in portfolio budgeting and financial feasibility constraints.

Selected Projects (AI / ML / Systems)

RAG4Konkur (Local RAG Study Assistant) 2026

- Built a local retrieval-augmented generation (RAG) system for study workflows: flashcard generation, concept review, and AnkiDroid export.
- Focused on practical usability, reproducible content generation, and offline-first study pipelines.

ML Engineering Preparation Projects (End-to-End Pipelines) 2023–present

- House-price ML pipeline: data preprocessing, feature engineering, model training, evaluation, and reporting.
- PyTorch CIFAR-10 pipeline: custom dataset loading, augmentation, checkpointing, and ResNet18 transfer learning.
- Model serving: deployed a trained model behind a FastAPI REST API; containerized deployment with Docker.
- Explored experiment tracking and deployment concepts (including MLflow-oriented ideas and CI/CD exposure).

Additional Applied / Exploration Projects 2023–present

- Robotic arm reinforcement learning simulation (exploration).
- Logistics heatmap decision-support tool; live stock candlestick/treemap visualization experiments.
- Workflow automation and AI tooling prototypes (LangChain-based apps; N8N-style orchestration).
- Full-stack utilities: spreadsheet web app; web dashboards and small decision-support interfaces.

Professional Experience

Project Management Officer (PMO), Iran Energy Industries Corporation (IEIC) 2023

- Served as senior deputy to the Head of Department; supported planning and coordination across project activities.
- Worked in project-oriented environments where budgeting, scheduling, and reporting constraints are central.

Project Control Specialist, Jahanpars Group 2019–2020

- First professional role during undergraduate studies; contributed to project control activities in an industrial setting (COVID-19 period).

Career Break (Market Re-entry; Technical Deepening) 2023–present

- Intentional shift from management-oriented work toward deep technical engineering for designing intelligent systems.
- Completed / progressed through full-stack and software engineering training (freeCodeCamp

tracks).

- Built multiple applied AI/ML engineering projects (pipelines, deployment, containerization, experimentation).

Technical Skills

Programming: Python; JavaScript; TypeScript; Markdown; LaTeX
Backend / APIs: FastAPI; Django; Express.js
Frontend: React.js
AI / ML: machine learning; deep learning (PyTorch); prompt engineering; LangChain; RAG
Data: EDA; preprocessing; feature engineering; visualization
MLOps / DevOps: Docker; model serving (API); deployment concepts; experiment tracking concepts
Automation / Workflow: N8N-style orchestration (as used in projects)

Certifications (Selected)

freeCodeCamp — Full Stack JavaScript Certifications (multiple tracks): Responsive Web Design; JavaScript Algorithms & Data Structures; Front End Libraries; Data Visualization; Relational Database; Back End Development and APIs.

Languages

Persian (Native); English (Professional working proficiency)